ID Document Forgery Detection

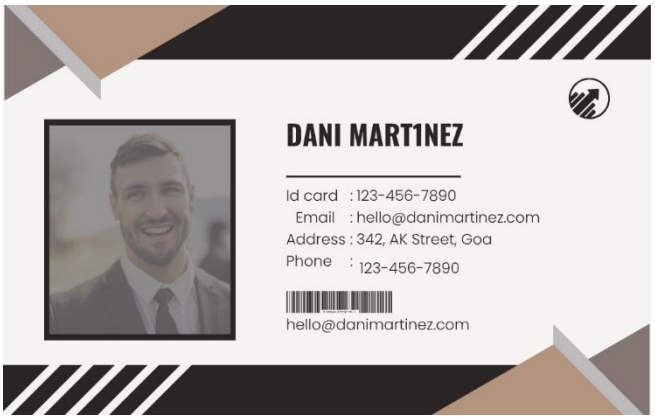
Sample input(s) along with corresponding output report(s)

Due to ethical and legal considerations, real-world forged or tampered ID documents were not used. Instead, controlled sample and synthetic test cases were used to simulate tampering scenarios.

1. Sample — Tampered ID

Input:

File **:** test_samples/tamp_ID.png
Decsription : Tampered ID Card



Output Summary:

- Document Type: ID Card
- Risk Level: CRITICAL
- Confidence: 90%
- Validity: LIKELY_VALID (The document follows a valid ID structure; however, detected anomalies indicate possible tampering.)

Key Findings:

- Name contains anomaly:
  → **“MART1NEZ” instead of “MARTINEZ”**

- Font inconsistency detected
- Missing standard security features
- Generic layout (template-like structure)

Tampering Indicators:

- Character substitution (I → 1) in name field
- Text field inconsistency
- Abnormal text density (possible overlay)
- High noise level (editing artifact)

Conclusion:

The system successfully detects subtle tampering in the identity field.
The use of a numeric character ('1') in place of a letter ('I') is a strong indicator of data manipulation, leading to a **CRITICAL risk classification**.

Full JSON Output : report_output_tamp_ID1.json :


```
{
  "document_type": "ID Card",
  "overall_risk_level": "CRITICAL",
  "confidence_score": 90,
```

"summary": "The document presents as a basic ID card but contains a critical anomaly in the name field, where a numeral '1' is used instead of the letter 'I' in 'MARTINEZ'. This, combined with the absence of standard security features, indicates a high likelihood of being an invalid or tampered document.",

"checks": [

{

"check_name": "Font Consistency",

"status": "FAIL",

"detail": "The name 'DANI MART1NEZ' uses a numeral '1' instead of the letter 'I', which is a significant inconsistency and suggests a typo or deliberate alteration. Other fonts appear consistent."

},

{

"check_name": "Alignment & Spacing",

"status": "PASS",

"detail": "Text fields and data elements appear to be properly aligned and spaced without obvious irregularities."

},

{

"check_name": "Photo Integrity",

"status": "PASS",

"detail": "The photo appears naturally embedded with consistent resolution and no obvious hard edges or lighting inconsistencies suggesting a swap."
},
{
"check_name": "Document Layout",
"status": "WARN",
"detail": "The overall layout is generic and lacks typical security elements found on official ID documents, such as holograms or complex background patterns. It resembles a basic template."
},
{
"check_name": "Print Quality & Artifacts",
"status": "PASS",
"detail": "The print quality appears consistent across the document with no significant compression artifacts, blurring, or inconsistent ink density in specific regions."
},
{
"check_name": "Security Features",
"status": "FAIL",
"detail": "No visible advanced security features such as microprint, guilloche patterns, or holograms are present. A

barcode is visible, but its content is not verifiable, and there is no Machine-Readable Zone (MRZ)."
},
{
"check_name": "Metadata & Lighting Coherence",
"status": "INCONCLUSIVE",
"detail": "Insufficient information to assess metadata or lighting coherence from the provided static image."
},
{
"check_name": "Text Field Tampering",
"status": "FAIL",
"detail": "The name 'MART1NEZ' clearly shows a numeral '1' instead of the letter 'I', which is a strong indicator of either a critical data entry error or deliberate text alteration. No other obvious background color differences or ghosting were observed."
}
],
"red_flags": [
"The name 'MART1NEZ' contains a numeral '1' instead of the letter 'I', indicating a likely typo or deliberate alteration.",
"Lack of standard security features (e.g., holograms, microprint, MRZ) common on official ID documents.",

"Generic design and layout, suggesting it might be a template or sample rather than an officially issued ID."
],
"positive_indicators": [
"The photo appears clear and well-integrated.",
"Overall text alignment and spacing are consistent (excluding the '1' anomaly)."
],
"analyst_notes": "The most significant issue is the '1' in the surname, which fundamentally compromises the integrity of the primary identifier. This document cannot be considered a valid ID due to this critical discrepancy and the absence of expected security features.",
"disclaimer": "This is an automated AI analysis for prototype/evaluation purposes only. It should not be used as sole evidence in legal or security decisions.",
"manual_analysis": {
"blur": 2089.805090890463,
"noise": 82.20401356598042,
"edges": 0.03366721927343819,
"color_variance": 17.328837708045963,
"compression": 0.05745110660791397,
"text_density": 0.3716071569542544
},

```
  "rule_based": {
    "status": "review",
    "issues": [
      "High noise level (possible editing)",
      "Abnormal text density (possible overlay)"
    ],
    "confidence": 0.38
  },
  "validity_status": "LIKELY_VALID",
  "metadata": {
    "filename": ".\\test_samples\\tamp_ID.png",
    "file_size_kb": 131.1,
    "media_type": "image/jpeg",
    "analyzed_at": "2026-04-01T10:24:00.821955Z",
    "model": "gemini-2.5-flash + OpenCV hybrid"
  }
}
```

2. Sample — Specimen ID (Slovak)

Input:

- File: test_samples/ID2.jpg
- Description: Specimen National ID (contains "Vzorka")

Output Summary:

- **Risk Level:** LOW_RISK
- **Validity:** SPECIMEN_DOCUMENT
- **Confidence:** 95%

Key Findings:

- Document is marked as “Vzorka” (Specimen)
- Contains placeholder data (e.g., dummy name, DOB)
- Proper alignment and layout
- Presence of security patterns (guilloche)

- No signs of tampering (no font mismatch, no editing artifacts)

Conclusion:

The document is structurally genuine and shows no signs of tampering.
However, it is clearly a **specimen/sample ID**, not a valid identification document.

Full JSON Output : report_output_ID1.json :

{

"document_type": "National ID",

"overall_risk_level": "LOW_RISK",

"confidence_score": 95,

"summary": "The document is identified as a Slovak National ID card. It is explicitly marked as a 'Specimen' (Vzorka) document, indicating it is not a valid identification document for a real person. No signs of digital tampering or physical alteration were detected on the specimen itself.",

"checks": [

{

"check_name": "Font Consistency",

"status": "PASS",

"detail": "All text fields exhibit consistent font styles and sizes, including the 'Specimen' markings and placeholder data."
},
{
"check_name": "Alignment & Spacing",
"status": "PASS",
"detail": "Text fields and data elements are properly aligned and spaced, consistent with a standard ID card layout."
},
{
"check_name": "Photo Integrity",
"status": "PASS",
"detail": "The grayscale photo appears naturally embedded with no hard edges, lighting inconsistencies, or resolution mismatches suggesting a swap."
},
{
"check_name": "Document Layout",
"status": "PASS",
"detail": "The overall layout, placement of elements, and background patterns are consistent with known Slovak National ID card conventions."

},
{
"check_name": "Print Quality & Artifacts",
"status": "PASS",
"detail": "Print quality is good with clear text and background patterns. Low compression artifacts (0.0434) and no inconsistent ink density or signs of re-digitization were observed."
},
{
"check_name": "Security Features",
"status": "PASS",
"detail": "Guilloche background patterns are clearly visible. The 'SK' logo and national emblem are present. While a specimen, the visual elements suggest proper security feature integration."
},
{
"check_name": "Metadata & Lighting Coherence",
"status": "PASS",
"detail": "Lighting and reflections across the document are consistent, indicating no compositing of different image parts."
},

{
"check_name": "Text Field Tampering",
"status": "PASS",
"detail": "No signs of text field tampering such as different background colors, altered font weights, ghosting, or erased text were detected."
}
],
"red_flags": [
"Document is explicitly marked as 'Specimen' (Vzorka) and is not a valid identification document.",
"Contains placeholder data (e.g., 'Specimen' for name, '11.11.1911' for date of birth, '111111/1111' for personal number)."
],
"positive_indicators": [
"Consistent font usage and sizing across all data fields.",
"Proper alignment and spacing of all textual elements.",
"Photo appears genuinely integrated into the document design.",
"Document layout and design are consistent with a genuine Slovak National ID card template.",
"High print quality with visible guilloche patterns and no significant digital artifacts."

],
"analyst_notes": "The document is a sample/specimen ID card, not intended for actual identification. The risk assessment pertains to the integrity of the specimen document itself, not its validity as a real ID for a person. No evidence of tampering with the specimen document was found.",
"disclaimer": "This is an automated AI analysis for prototype/evaluation purposes only. It should not be used as sole evidence in legal or security decisions.",
"manual_analysis": {
"blur": 804.3244143285206,
"noise": 42.61903214338967,
"edges": 0.10896485108428716,
"color_variance": 40.07565954878675,
"compression": 0.043434254825115204,
"text_density": 0.2946024354229645
},
"rule_based": {
"status": "likely_genuine",
"issues": [
"Moderate noise detected",
"Abnormal text density (possible overlay)"
],

```
    "confidence": 0.25
  },
  "validity_status": "SPECIMEN_DOCUMENT",
  "metadata": {
    "filename": ".\\test_samples\\ID2.jpg",
    "file_size_kb": 215.7,
    "media_type": "image/jpeg",
    "analyzed_at": "2026-04-01T08:27:18.911660Z",
    "model": "gemini-2.5-flash + OpenCV hybrid"
  }
}
```

3. Sample — Driving Licence Template

Input:

- File: test_samples/ID3.jpg
- Description: Specimen National ID (contains "Vzorka")

Output Summary:

- **Risk Level:** LOW_RISK
- **Validity:** NOT_VALID_DOCUMENT
- **Confidence:** 95%

Key Findings:

- Uses placeholder data (e.g., “NAME SURNAME”)
- Contains generic avatar photo
- Lacks real security features (no microprint, etc.)

- Looks like an illustration/template

Conclusion:

The document is not tampered, but it is not a real ID. It is a generic sample/illustration used for demonstration purposes.

Full JSON Output : report_output_ID2.json :

{
"document_type": "Driver License",
"overall_risk_level": "LOW_RISK",
"confidence_score": 95,
"summary": "The document is an illustration of a driver's license, featuring generic placeholder data and an avatar photo. It shows no signs of digital tampering or manipulation, but it is not a valid identification document.",
"checks": [
{
"check_name": "Font Consistency",
"status": "PASS",
"detail": "Fonts appear consistent across all text fields, with appropriate variations for headings and data."

},
{
"check_name": "Alignment & Spacing",
"status": "PASS",
"detail": "Text fields and data elements are properly aligned and spaced, showing no irregularities."
},
{
"check_name": "Photo Integrity",
"status": "PASS",
"detail": "The photo is a generic avatar, consistent with the illustrative nature of the document. There are no hard edges, lighting inconsistencies, or resolution mismatches suggesting a swapped photo."
},
{
"check_name": "Document Layout",
"status": "PASS",
"detail": "The overall layout matches a typical driver's license design, with proper placement of data fields and a generic security element (hologram)."
},
{

"check_name": "Print Quality & Artifacts",
"status": "PASS",
"detail": "The image exhibits good print quality with no noticeable compression artifacts, blurring in specific regions, or inconsistent ink/print density. OpenCV signals are consistent with a clean digital image."
},
{
"check_name": "Security Features",
"status": "WARN",
"detail": "A generic 'VERIFIED' hologram and a barcode are present. However, advanced security features like microprint or complex guilloche patterns are not discernible or are absent, which is expected for an illustrative sample."
},
{
"check_name": "Metadata & Lighting Coherence",
"status": "PASS",
"detail": "As an illustration, lighting is uniform and consistent across the document, with no inconsistent shadows or reflections."
},
{
"check_name": "Text Field Tampering",

"status": "PASS",

"detail": "No signs of tampering were detected in text fields, such as different background colors, altered font weights, or ghosting."

}

],

"red_flags": [

"Document is a generic illustration/sample and not a valid identification document."

],

"positive_indicators": [

"Clear and consistent document layout",

"Uniform font usage",

"Presence of a generic 'VERIFIED' hologram and barcode",

"No visible signs of digital manipulation or tampering"

],

"analyst_notes": "The document is clearly an illustrative sample or template for a driver's license, indicated by placeholder text like 'NAME SURNAME' and a generic avatar. While it is not a valid ID, there are no forensic indicators of tampering or forgery on the illustration itself.",

"disclaimer": "This is an automated AI analysis for prototype/evaluation purposes only. It should not be used as sole evidence in legal or security decisions.",

"manual_analysis": {
  "blur": 2219.729392624729,
  "noise": 57.89940599143125,
  "edges": 0.04291998071824536,
  "color_variance": 56.739674433009455,
  "compression": 0.06033976376056671,
  "text_density": 0.14740298867196916
},
"rule_based": {
  "status": "likely_genuine",
  "issues": [
    "Moderate noise detected"
  ],
  "confidence": 0.12
},
"validity_status": "NOT_VALID_DOCUMENT",
"metadata": {
  "filename": ".\\test_samples\\ID3.jpg",
  "file_size_kb": 25.8,
  "media_type": "image/jpeg",
  "analyzed_at": "2026-04-01T08:35:48.924682Z",
  "model": "gemini-2.5-flash + OpenCV hybrid"

```
  }
}
```